Ogbonna Arinze

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SUMMARY

Senior Frontend Engineer with 8 years building delightful, performant **React** + **TypeScript** single-page apps for technical users. Has shipped a **Next.js** AI chat widget at Genentech, a real-time SSE-streamed chat platform at OurNigeria, and a clinical trials Chrome extension migrated from Vue.js. Strong feel for HTTP, REST, and modern API patterns from years of fullstack work. Combines product taste, performance instinct, and written-first communication, with a multi-year track record of side projects and open-source-style ownership across the stack.

CORE COMPETENCIES

- **Core Stack:** TypeScript (advanced generics, utility types), React, Next.js, TailwindCSS, modern CSS, semantic HTML.
- **HTTP and APIs:** Deep REST experience (NestJS, NodeJS), GraphQL, JWT auth flows with HTTP-only cookies, SSE streaming, WebSockets, EventSource API, request caching.
- **Complex UI Patterns:** Real-time chat surfaces, streaming responses, drag-and-drop friendly architectures, keyboard-first interactions, design systems on Radix UI.
- **Performance:** Lighthouse, Core Web Vitals, code splitting, React Profiler, memoization, bundle analysis (webpack-bundle-analyzer, rollup-plugin-visualizer).
- **Testing and Automation:** Playwright E2E in CI with Allure reporting, Jest, Cypress, Storybook, GitHub Actions.
- **AI/LLM Integration:** OpenAI API, LangChain, Vercel AI SDK, Mastra multi-agent framework, RAG pipelines.
- **Communication:** Written-first specs and code review, fully distributed since 2020.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)
Frontend Engineer — *May 2023 - Present*

- Co-developed a full-stack **Next.js** chat widget powered by a Retrieval-Augmented Generation system using OpenAI GPT models, **LangChain**, **Vercel AI SDK**, and **Chroma**, embeddable on external sites with delightful streaming UX for technical users.
- Spearheaded the redesign of the main clinical trial data product, migrating a **Vue.js** SPA to a **React** Chrome extension with **TypeScript**, owning UX, architecture, and rollout.
- Architected a reusable design system in **React** and **Radix UI** with **Tailwind CSS**, raising UI consistency, accessibility (WCAG/ARIA), and developer velocity across teams.
- Implemented **JWT-based** auth flows on the frontend with HTTP-only cookies, protected routes, RBAC-driven UI, and seamless token refresh, working closely with backend on the contract.
- Optimized application performance through lazy loading, code splitting, image optimization, memoization, and CDN delivery, profiled bottlenecks with **Lighthouse** and Chrome DevTools.
- Introduced **Redux** for client state and **React Query** for server-state caching, refactored class components to functional, tuned the build pipeline.
- Integrated **Playwright** E2E into **GitHub Actions**, published **Allure Reports** on GitHub Pages, and pushed Slack notifications with deep links to test results.
- Maintained the existing **Vue.js** codebase in parallel, fixing bugs and shipping features without regression during the migration.
- Mentored two junior developers (interns) through written-first reviews and pair programming.

Descasio — Lagos, Nigeria (Hybrid)
Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of a greenfield **React** + **TypeScript** SPA, achieving a **25% boost in user engagement** and accelerating delivery by **30%**.
- Profiled render bottlenecks with **React Profiler** and **Lighthouse**, cut bundle size **30-40%** using rollup-plugin-visualizer and webpack-bundle-analyzer, achieving **35% faster** initial load.
- Drove a **40% improvement** in runtime performance by reworking caching strategies across the React application.
- Documented React components with **Storybook** and introduced **Jest** + **Enzyme** testing, leading to a **20% drop** in bug reports.
- Established team engineering standards via **ESLint**, **Prettier**, **Husky**, **Lint-staged**, and **GitHub Actions**.
- Mentored junior engineers on frontend best practices.

ActivEdge Technologies — Lagos, Nigeria
Senior React Developer — *Oct 2019 - Sep 2021*

- Built a high-volume bank Settlement Solution in a team of 3, handling thousands to millions of transactions daily and boosting settlement rates by **80%**.

- Adopted the native **EventSource API** for real-time data streaming over socket.io, improving latency and reducing dependencies.
- Developed payment modules (MasterCard, VISA, Express-cash, E-commerce) with **React**, **Redux**, **RxJS**, and **Axios**.
- Built a real-time account reconciliation tool, lifting daily reconciliation by **60%**.
- Built dynamic data visualization dashboards with **react-chartjs**, **Redux Toolkit**, and **Ant Design**.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS** RESTful APIs and translated business requirements into efficient **MongoDB** models.
- Converted Figma designs into reusable **React** components for a consistent UI.
- Engineered a CI/CD pipeline for automated API deployment to Heroku and AWS production servers.
- Mentored two interns through frontend and backend project work.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Built **SureEdu**, a school management portal, in a team of 5 across staff, student, and guardian modules.
- Designed **MongoDB** schemas and implemented server-side logic in **NodeJS** with REST endpoints.
- Authored use cases, translated them into database models, and wrote feature documentation per application.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget transparency platform with multi-agent LLM pipeline analyzing 700+ documents and 708K+ vector embeddings.

- Architected the full platform as an **Nx monorepo** of 6 apps (NestJS API, NestJS ingestion pipeline, three Next.js frontends, Remotion video service) orchestrated with **pnpm**.
- Implemented **SSE streaming** for real-time chat with keepalive heartbeats every 15s, token-budgeted context windows, conversation summarization, and memory extraction.
- Built a hybrid **RAG** pipeline (pgvector + BM25 + Reciprocal Rank Fusion + Cohere reranking) over 708K+ Voyage AI embeddings.
- Built citizen-facing **Next.js** app with **React 19**, **TailwindCSS v4**, dark mode, URL state via nuqs, and **Recharts** visualizations auto-generated from AI responses.
- Configured CI/CD with **GitHub Actions**, **Playwright** E2E across 4 projects, and Docker Compose production deployment with multi-stage Dockerfiles.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*